Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming 1st part–February 9th, 2026

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

A. Please show an example where idling processors is unavoidable, and explain how it affects efficiency. (1)

B. A programmer claims: "If my algorithm runs on EREW, it will also run on CREW and CRCW without changes." Is this always true? Explain. (1)

C. Suppose you have a PRAM algorithm that heavily relies on random access to many distinct shared cells per step. Based on the lemma about the scaling of the performance of a PRAM with reduced memory footprint, would you expect it to degrade more (or less) than an algorithm that mostly reuses a small working set? Motivate the answer. (1)

D. You observe near-linear speedup when increasing processors. Is this evidence supporting Amdahl's model or Gustafson's model? Explain why. (1)

Exercise n. 2

A. Both RAND-SELECT and SELECT find the i-th order statistic, but only SELECT guarantees worst-case time complexity. What specific pivot-selection mechanism does SELECT use to limit the size of the largest recursive subproblem, and how does that guarantee lead to the worst-case bound (i.e., what fraction of elements is discarded each step)? (2)

B. Explain when the quick sort algorithm falls in the best case or in the worst case recurrence. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

C. Describe how the insert for the skip-list data structure works and discuss its complexity. (1)

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Exercise n. 3

A. What kind of spatial complexity does the longest common subsequence problem have? (1)

B. Treap required all priorities to be pairwise distinct. How can this property be satisfied? (1)

C. Karger's min-cut algorithm contracts uniformly random edges until only two supernodes remain and outputs the cut defined by them. In a few sentences, explain (i) what can cause a single run to return a non-minimum cut, and (ii) why and how running the same algorithm many times changes the reliability of the result. (2)

Exercise n. 4

A. Explain the different roles of the unique table and the computed table. Give a scenario where the unique table alone is insufficient to avoid exponential time, and the computed table fixes it. (2)

B. Explain why the self-organizing list problem is naturally online, and give a concrete access sequence where having future knowledge (offline) allows strictly lower cost than any online strategy. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**